

BABUSHKIN, Russian Federation

WMO: 308220

Lat: 51.708N

Lon: 105.852E

Elev: 1534

StdP: 13.90

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.2	-10.1	-20.4	1.9	-14.3	-15.4	2.5	-8.9	23.5	3.9	22.2	4.9	3.7	240	0.694

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	72.9	62.0	69.9	61.2	67.6	60.3	64.5	70.8	62.8	68.0	61.4	66.3	5.7	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.0	88.0	67.6	60.5	83.4	65.9	59.1	79.1	64.6	30.5	71.0	29.3	68.1	28.2	66.4	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	21.3	18.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-17.9	67.8	5.1	4.2	-21.6	70.8	-24.6	73.2	-27.4	75.6	-31.2	78.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.4	8.2	5.9	18.8	33.9	43.8	53.3	60.7	59.6	49.4	36.9	23.9	16.5
	DBStd	19.93	9.19	9.68	10.17	6.10	5.75	4.97	4.58	4.42	5.52	6.65	8.27	7.62
	HDD50	6513	1295	1235	967	483	205	22	0	1	77	408	782	1038
	HDD65	11183	1760	1655	1432	932	657	350	146	175	470	871	1232	1503
	CDD50	824	0	0	0	0	13	122	333	297	57	2	0	0
	CDD65	19	0	0	0	0	0	0	13	6	0	0	0	0
	CDH74	62	0	0	0	0	2	2	39	17	2	0	0	0
	CDH80	7	0	0	0	0	0	0	6	1	0	0	0	0
	WSAvg	7.4	7.9	6.1	6.3	7.2	7.7	6.8	5.8	6.2	7.6	7.8	9.3	10.1
	PrecAvg	20.6	0.7	0.3	0.4	1.0	1.3	2.7	4.4	3.5	2.8	1.1	1.0	1.1
(15) Precipitation	PrecMax	28.3	1.6	0.9	1.3	2.3	2.6	8.1	12.0	8.5	7.4	3.3	2.1	2.5
	PrecMin	14.8	0.1	0.0	0.1	0.0	0.1	0.2	0.6	0.7	0.7	0.2	0.2	0.2
	PrecStd	3.6	0.4	0.2	0.3	0.6	0.6	1.8	2.6	1.7	1.5	0.7	0.4	0.6
	PrecStd	3.6	0.4	0.2	0.3	0.6	0.6	1.8	2.6	1.7	1.5	0.7	0.4	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	32.2	31.7	46.4	58.6	69.6	72.5	78.3	75.8	69.0	58.3	44.7	36.1	36.1
	0.4% MCWB	29.3	26.7	36.3	43.4	51.6	58.8	65.3	63.2	57.4	47.4	38.3	30.8	30.8
	2% DB	27.9	27.4	40.2	50.7	61.8	68.0	73.8	71.9	63.6	52.1	41.3	32.6	32.6
	2% MCWB	25.3	24.0	33.7	40.5	47.9	57.8	63.9	61.4	54.4	44.7	35.8	28.7	28.7
	5% DB	24.6	23.7	36.3	46.4	57.4	64.7	70.7	69.1	60.4	49.0	37.8	30.1	30.1
	5% MCWB	22.0	20.8	31.4	38.2	46.6	55.9	62.7	61.2	53.3	42.7	33.3	27.0	27.0
	10% DB	20.7	20.0	33.4	43.2	53.9	62.0	68.3	66.8	57.8	46.4	35.2	27.3	27.3
	10% MCWB	18.5	17.8	29.3	36.5	44.6	54.7	61.5	60.3	52.0	41.0	30.9	24.4	24.4
	0.4% WB	29.4	27.5	37.5	45.2	53.8	62.3	67.0	65.6	60.2	48.2	39.4	31.4	31.4
	0.4% MCDB	32.0	30.9	43.2	55.4	65.7	67.9	73.2	71.4	65.7	56.2	44.2	34.9	34.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	25.7	24.2	34.5	41.5	50.0	58.9	65.4	63.5	56.6	45.8	36.3	29.2	29.2
	2% MCDB	28.1	27.4	39.8	49.6	58.0	65.3	71.4	69.0	61.7	51.7	40.4	32.3	32.3
	5% WB	22.2	20.9	31.9	38.8	47.6	57.0	63.9	62.3	54.3	43.5	33.6	27.0	27.0
	5% MCDB	24.4	23.6	35.8	45.5	56.0	63.0	69.9	67.2	58.7	48.2	37.3	30.0	30.0
	10% WB	18.7	17.8	29.5	36.8	45.5	55.4	62.3	61.0	52.5	41.3	31.4	24.5	24.5
	10% MCDB	20.6	20.0	33.2	42.6	53.4	61.0	67.3	65.9	57.0	45.9	35.0	27.3	27.3
(35) Mean Daily Temperature Range	MDBR	12.6	15.0	15.8	13.4	14.0	13.1	11.1	10.7	11.8	11.6	11.0	10.6	10.6
	5% DB	14.0	17.9	19.1	19.9	21.7	18.0	15.7	15.1	16.4	15.6	12.5	12.3	12.3
	5% WB	12.1	15.4	13.4	11.4	11.3	10.1	8.3	8.1	9.8	11.0	9.8	10.5	10.5
	MCDBR	13.5	17.8	16.7	18.7	19.1	15.2	13.3	12.8	14.5	14.6	12.1	11.9	11.9
	MCWBR	11.9	15.3	12.1	11.3	11.0	9.6	7.8	7.7	9.5	10.9	10.0	10.5	10.5
	MCWBR	11.9	15.3	12.1	11.3	11.0	9.6	7.8	7.7	9.5	10.9	10.0	10.5	10.5
(40) Clear-Sky Solar Irradiance	taub	0.269	0.295	0.331	0.378	0.442	0.445	0.434	0.394	0.340	0.316	0.288	0.270	0.270
	taud	2.120	2.104	2.146	2.236	2.089	2.199	2.278	2.375	2.473	2.432	2.207	2.090	2.090
	Ebn at Noon	229	257	271	272	258	258	259	265	269	253	220	203	203
	Edh at Noon	23	32	38	39	48	43	40	34	28	23	22	21	21
(44) All-Sky Solar Radiation	RadAvg	406	743	1175	1506	1801	1878	1739	1466	1135	711	403	285	285
	RadStd	20	36	59	95	98	109	114	83	109	50	27	21	21

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page